

Boundary and corner charts in low-dimensional inverse problems

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Abstract

Alberti, Arroyo and Santacesaria ask about extending their low-dimensional-manifold stability theory to settings with boundary points, for instance degenerate polygonal parameter regimes. We prove a fixed-dimensional extension: their compact-set Holder and Lipschitz stability results remain valid for manifolds whose charts take values in relatively open subsets of closed convex finite-dimensional models, such as half-spaces and orthants, provided the charted forward maps admit C^1 Euclidean extensions with injective differentials. The same replacement gives the finite-measurement $Q_N F$ stability theorem. This addresses the boundary/corner part of the question, but not variable-dimensional stratified degenerations.

1 Source Question

The source paper proves abstract stability results for nonlinear inverse problems

$$F(x) = y$$

when the unknown x lies in a compact subset K of a finite-dimensional differentiable manifold $M \subset X$, with X, Y Banach spaces. Its main theorem says, roughly, that if M is Holder in X , $F \in C^1(M, Y)$, F is injective, and dF_x is injective on every tangent space, then F is Holder stable on K . A finite-measurement theorem gives analogous stability for $Q_N F$.

In the conclusion, the authors list the following open direction:

- learning may be used to learn some of the parameters of the network, like those related to the manifold M which, as mentioned before, may be unknown.
- The manifolds considered in this work are without boundary and, by definition, of fixed dimension. However, there are situations where boundary points may be of interest, in order to cover, for instance, the case of degenerate polygons, for which the number of parameters vary. Extending the current theory to this generalised setting would enlarge the range of applicability of these results.
- Another key assumption of the main results is the injectivity of the differential of F . However, in some cases, the differential of F may not be injective. It would be interesting to investigate whether a workaround may be found by using higher order derivatives of F , as in certain higher order

The result below gives a precise partial answer: fixed-dimensional boundary and corner charts are harmless for the abstract stability mechanism.

2 Definitions

Let

$$C_{n,m} = [0, \infty)^m \times \mathbb{R}^{n-m} \subset \mathbb{R}^n, \quad 0 \leq m \leq n,$$

and call any translate and linear image of such a set a closed corner model. The proof works verbatim for any closed convex $C \subset \mathbb{R}^n$ with nonempty relative interior, but the displayed models are the relevant boundary/corner cases.

Definition 1. Let X be a Banach space. A set $M \subset X$ is an n -dimensional corner manifold in X if it has an atlas $\{(U_i, \varphi_i)\}_{i \in I}$ such that:

- U_i is relatively open in M ;
- $\varphi_i : U_i \rightarrow A_i$ is a homeomorphism onto a relatively open subset A_i of a closed corner model $C_i \subset \mathbb{R}^n$;
- the transition maps admit C^1 extensions to open Euclidean neighbourhoods of their domains.

We say that M is α -Holder in X , with $0 < \alpha \leq 1$, if there is $\ell > 0$ such that

$$\|x - y\|_X \leq \ell^{-1} |\varphi_i(x) - \varphi_i(y)|^\alpha, \quad x, y \in U_i.$$

The case $\alpha = 1$ is called Lipschitz in X .

Definition 2. For a corner manifold M , a map $F : M \rightarrow Y$ is C^1 in the extendable chart sense if every charted map

$$F_i := F \circ \varphi_i^{-1} : A_i \rightarrow Y$$

admits a C^1 extension $\tilde{F}_i$ to an open neighbourhood of A_i in $\mathbb{R}^n$. We write dF_x for $D\tilde{F}_i(\varphi_i(x))$, up to the usual change of coordinates. Injectivity of dF_x means injectivity as a linear map $\mathbb{R}^n \rightarrow Y$.

3 Idea of the Proof

The source proof is local. Compactness first reduces the problem to pairs of nearby points lying in a single chart. Inside a boundaryless chart, the source uses the line segment between two close coordinate points and integrates the derivative along that segment.

For a boundary or corner chart, the coordinate domain is no longer open in all of $\mathbb{R}^n$. But if it is relatively open in a closed convex model, short line segments between points of a compact subset still stay in the model and, after shrinking the distance scale, stay inside the relative chart domain. Since the charted forward map has a C^1 extension to an ordinary open Euclidean neighbourhood, the same fundamental theorem of calculus argument applies. This is the whole mechanism.

4 Finite-Dimensional Relative Lemma

Lemma 1. Let $C \subset \mathbb{R}^n$ be closed and convex, let $A \subset C$ be relatively open, let $K \subset A$ be compact, and let Y be a Banach space. Suppose $G : A \rightarrow Y$ is injective and admits a C^1 extension $\tilde{G}$ to an open neighbourhood $O \subset \mathbb{R}^n$ of A . If $D\tilde{G}(a) : \mathbb{R}^n \rightarrow Y$ is injective for every $a \in A$, then there is $L > 0$ such that

$$\|u - v\| \leq L \|G(u) - G(v)\|_Y, \quad u, v \in K.$$

Proof. Large distances are controlled by compactness and injectivity. Namely, for every $\rho > 0$,

$$\min\{\|G(u) - G(v)\|_Y : u, v \in K, |u - v| \geq \rho\} > 0$$

unless the set is empty.

It remains to handle nearby points. Because K is compact in the relatively open set $A \subset C$, there is $\rho_0 > 0$ such that

$$\widehat{K} := \{z \in C : \text{dist}(z, K) \leq \rho_0\}$$

is a compact subset of $A \cap O$. If $u, v \in K$ and $|u - v| \leq \rho_0$, then the segment $[u, v]$ is contained in $\widehat{K}$, since C is convex.

By compactness of $K \times S^{n-1}$ and injectivity of the derivative,

$$c := \frac{1}{2} \inf_{u \in K, |e|=1} \|D\tilde{G}(u)e\|_Y > 0.$$

Since $D\tilde{G}$ is uniformly continuous on the compact set $\widehat{K}$, choose $0 < \rho \leq \rho_0$ so that

$$\|D\tilde{G}(z) - D\tilde{G}(u)\|_{\mathcal{L}(\mathbb{R}^n, Y)} \leq c$$

whenever $u \in K$, $z \in \widehat{K}$, and $|z - u| \leq \rho$.

For $u, v \in K$ with $0 < |u - v| \leq \rho$, write $\gamma(t) = u + t(v - u)$. Then

$$\begin{aligned} \|G(v) - G(u)\|_Y &\geq \|D\tilde{G}(u)(v - u)\|_Y - \int_0^1 \|(D\tilde{G}(\gamma(t)) - D\tilde{G}(u))(v - u)\|_Y dt \\ &\geq (2c - c)|u - v| = c|u - v|. \end{aligned}$$

Combining this with the large-distance estimate gives the claimed Lipschitz inverse bound on all of K . $\square$

5 Boundary/Corner Stability

Theorem 1. *Let X, Y be Banach spaces, $0 < \alpha \leq 1$, let $M \subset X$ be an n -dimensional corner manifold which is α -Holder in X , and let $K \subset M$ be compact. Let $F : M \rightarrow Y$ be C^1 in the extendable chart sense. Assume:*

- *F is injective on M ;*
- *$dF_x : \mathbb{R}^n \rightarrow Y$ is injective for every $x \in M$.*

Then there is $C_K > 0$ such that

$$\|x - y\|_X \leq C_K \|F(x) - F(y)\|_Y^\alpha, \quad x, y \in K.$$

Proof. The one-chart compactness lemma from the source uses only that the U_i 's are relatively open in M and cover the compact set K . Therefore there are $\delta > 0$ and compact sets $K_1, \dots, K_N \subset K$ covering K such that, for each j , some chart $U_{i(j)}$ contains

$$\overline{B_X(K_j, \delta)} \cap M.$$

If $\|x - y\|_X \geq \delta$, compactness of K and injectivity of F give the desired estimate after enlarging the constant. Suppose now that $\|x - y\|_X < \delta$. Choose j with $x \in K_j$; then $y \in U_{i(j)}$. Set $\varphi = \varphi_{i(j)}$, $A = \varphi(U_{i(j)})$, and $G = F \circ \varphi^{-1}$. The compact coordinate set

$$H_j = \varphi(K \cap \overline{B_X(K_j, \delta)}) \subset A$$

is contained in a relatively open subset of a closed convex model. By Lemma 1, there is $L_j > 0$ such that

$$|\varphi(x) - \varphi(y)| \leq L_j \|F(x) - F(y)\|_Y$$

for all such x, y . The α -Holder chart condition gives

$$\|x - y\|_X \leq \ell^{-1} |\varphi(x) - \varphi(y)|^\alpha \leq \ell^{-1} L_j^\alpha \|F(x) - F(y)\|_Y^\alpha.$$

Taking the maximum over finitely many charts, and combining with the large-distance constant, proves the theorem. $\square$

Theorem 2. *Let M, K, F be as in Theorem 1 with $\alpha = 1$. Let $Q_N : Y \rightarrow Y$ satisfy the source Hypothesis Q_N : there is a subspace $\tilde{Y} \subset Y$ and $D > 0$ such that $Q_N|_{\tilde{Y}}$ are uniformly bounded and converge strongly to the identity on $\tilde{Y}$. Assume additionally that*

$$F(x) - F(y) \in \tilde{Y} \quad (x, y \in K), \quad \text{ran}(dF_x) \subset \tilde{Y} \quad (x \in M).$$

Then there are $C > 0$ and $N_0 \in \mathbb{N}$ such that

$$\|x - y\|_X \leq C \|Q_N F(x) - Q_N F(y)\|_Y, \quad x, y \in K, \quad N \geq N_0.$$

Proof. The proof is the source proof of Theorem QNF -stability with Lemma 1 replacing the open-domain finite-dimensional lemma. We spell out the only point where the boundary matters.

After the same finite chart reduction used in Theorem 1, nearby $x, y \in K$ are represented by $u = \varphi(x)$ and $v = \varphi(y)$ in a compact set $H_j \subset A$, where A is relatively open in a closed convex model. As in Lemma 1, short coordinate segments $[u, v]$ remain in a compact $\hat{H}_j \subset A$. Along those segments,

$$Q_N G(v) - Q_N G(u) = \int_0^1 Q_N D\tilde{G}(u + t(v - u))(v - u) dt.$$

The source argument estimates this integral by comparing it with the already stable $G(v) - G(u)$. The two required uniform convergence statements are on compact sets:

$$\sup_{a, b \in H_j} \|(I - Q_N)(G(a) - G(b))\|_Y \rightarrow 0$$

and

$$\sup_{z \in \hat{H}_j} \|(I - Q_N)D\tilde{G}(z)\|_{\mathcal{L}(\mathbb{R}^n, Y)} \rightarrow 0.$$

They follow from compactness of $G(H_j) - G(H_j) \subset \tilde{Y}$, compactness of $D\tilde{G}(\hat{H}_j)(B_{\mathbb{R}^n}) \subset \tilde{Y}$, and the strong convergence of the uniformly bounded maps $Q_N|_{\tilde{Y}}$. Thus the finite-dimensional estimate for $Q_N G$ holds in every chart for all sufficiently large N . The source's large-distance compactness argument handles pairs with $\|x - y\|_X \geq \delta$, and the finitely many chart constants can be maximized. This gives the displayed estimate. $\square$

6 Scope and Limitations

Theorems 1 and 2 cover fixed-dimensional manifolds with boundary and corners, including parameter domains locally modelled on orthants. This is enough for many constrained parameter families in which, for example, a radius or contrast parameter may reach an endpoint while the number of parameters stays fixed.

The result does not cover genuinely variable-dimensional degeneracies. If a triangle is allowed to collapse to an edge and then to a point, the natural parameter space is stratified rather than a single n -dimensional corner manifold with injective n -dimensional differential on every stratum. Handling that case would require either the source’s finite-union theorem applied to a controlled finite stratification, when available, or new compatibility estimates between strata.

7 Novelty Check

The run indexes were searched for arXiv:2009.00574, the title, “manifold with boundary”, “manifold with corners”, “degenerate polygons”, and “M-RIP”. No prior packet or attempt for this target or route was found. Web searches for exact phrases around the source title plus “manifolds with boundary”, “degenerate polygons”, and “M-RIP manifold with boundary” did not locate a separate later answer.

8 References

- G. S. Alberti, A. Arroyo, and M. Santacesaria, *Inverse problems on low-dimensional manifolds*, arXiv:2009.00574.